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Computer Science PhD Applicant, Fall 2025

From an early age, I have been captivated by the fields of computer science and geography. The idea of intersecting these disciplines to address complex, real-world challenges has always intrigued me. My curiosity deepened as I began to wonder about the fundamental role of mathematics in these fields—specifically, why math is so essential and how it underpins the coding process used to solve complex problems. This curiosity led me to explore the fields of pattern recognition, machine learning (ML), and image processing. I remained committed to this, learning courses from different online sources, and this pursuit has significantly shaped my journey in the field of ML and remote sensing.

Research Journey During my undergraduate studies, my thesis under Prof. Dr. Amin Ahsan Ali involved using deep learning to analyze remote sensing data for Land Use and Land Cover (LULC) classification and segmentation. This project presented several significant challenges, including data unavailability, imbalanced datasets, cloud coverage issues, and limited access to high-performance computational resources. We attempted downsampling the images to apply deep learning. However, this approach resulted in significant information loss and unsatisfactory accuracy. Understanding the need for a better solution, we developed a non-overlapping grid-based approach. My contribution included dividing the large images into smaller segments while meticulously tracking their positions for accurate reconstruction. We evaluated this approach with the DL model for segmentation and achieved satisfactory accuracy by analyzing confusion matrices and measuring the overlap between predicted and actual segments. This experience strengthened my technical expertise and reinforced my passion for addressing real-world challenges through innovative DL solutions.

Post-Graduate Research and Collaboration After graduation, I joined the CCDS lab, collaborating with Prof. Dr. Ryosuke Shibasaki *(University of Tokyo)*, the Data and Design Lab *(University of Dhaka)*, and the United Nations University. I contributed to developing a novel classification framework that classified urban areas based on architectural forms and levels of urbanization, leveraging Google's DeepLabV3+ model. Leading a team of 24 students, I oversaw dataset creation, annotation, and validation using GIS tools such as eCognition. This project continued to five additional cities—Mumbai, Lima, Jakarta, Guangzhou, and Nairobi—to compare complexities across developed, underdeveloped, and developing urban regions. I tailored datasets for various deep learning models, conducted training and testing, and evaluated their performance in each city. Our method effectively distinguishes between formal and informal settlements, providing valuable insights for urban planning and policy-making. This work led to a publication in the *Sensors Journal, MDPI*.

Disaster Classification Project and BD-Sat During my tenure at CCDS, I also contributed to a disaster classification project accepted at *ICPR*. We curated a dataset of 13,720 manually annotated images covering various disasters, including fire, water, and land, alongside images of damaged infrastructure and humans. I developed an interface to analyze the attention model on new test data and built a prototype using OpenCV, an open-source computer vision library, to manually identify and classify disasters. Additionally, I contributed to developing BD-Sat, a high-resolution dataset for LULC analysis, which is under review, enabling accurate training of deep learning models despite the scarcity of annotated data in developing countries. My specific contributions to this project involved rigorous data preprocessing, modifying the neural network model's architecture, and analyzing and visualizing results despite the challenges posed by limited high-resolution data.

Professional Experience Over the past years, I have expanded my expertise through professional work, contributing to innovative applications that involve complex image analysis and transformation. I led a team to develop an innovative system for the apparel industry, which allowed users to visualize how clothing would look on different body types. The project involved addressing challenges such as varying skin tones, clothing styles, diverse body shapes, and different poses. We needed to create a solution that would seamlessly integrate all these factors to provide realistic and accurate visual representations.

My contributions included sourcing and preparing data through web scraping (automatically gathering information from websites) and extensive preprocessing, which involved tasks such as removing unnecessary backgrounds, estimating body poses, and segmenting body parts to ensure high-quality input for the model. I also led the development and deployment of the model, ensuring that it could adapt to various clothing types, sizes, and user characteristics. Additionally, I guided the team in refining the model's performance and deployed it on a server to make it accessible for real-time use. This project not only enhanced my technical skills but also strengthened my ability to collaborate, lead a team, and drive a complex project toward successful completion.

Teaching and Mentorship Beyond research, I served as a Teaching Assistant for courses: introduction to programming, data structures, and numerical methods. As a research assistant, I mentored junior lab members, guiding them through thesis works and fostering collaboration.

Future Directions Looking ahead, I am passionate about leveraging artificial intelligence across diverse domains, such as environmental monitoring, urban planning, disaster management, and other interdisciplinary areas. By integrating AI with domain-specific knowledge, I aim to drive innovation and create impactful solutions.

I believe the University of Illinois, with its interdisciplinary research and distinguished faculty, offers an ideal platform to achieve these goals. I am particularly interested in working with Professors Alexander Schwing, Svetlana Lazebnik, and Kaiyu Guan. Professor Schwing's and Professor Lazebnik's expertise in machine learning and computer vision—particularly in object detection, segmentation, generative modeling, and human pose estimation—aligns closely with my prior work in deep learning for segmentation, generative modeling, and keypoint detection. Dr. Lazebnik's work on virtual try-on systems resonates deeply with my prior project in this domain, where I developed a novel system enabling users to visualize how clothing would look on different body types. Collaborating with her would not only enable me to further refine these methodologies but also explore innovative applications of computer vision, bridging academic research with impactful real-world solutions. Similarly, Professor Guan's research on applying remote sensing and computational methods to agricultural sustainability complements my experience in remote sensing and LULC analysis.

Collaborating with these professors would allow me to contribute meaningfully while advancing my expertise to tackle global challenges using AI and data-driven solutions. After graduate school, I hope to work in an academic setting as well as in a research organization.